

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Pharm. Examination

University Theory Examination April/May 2015

PH 312 Biopharmaceuticals

Date: 23.04.2015, Thursday Time: 10:00a.m to 01:00p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1 Attempt any TWO of the following.**
- | | | |
|---|---|----|
| A | Describe the production and recovery of the penicillin by fermentation process with suitable flow sheet. | 10 |
| B | Enumerate upstream and downstream processes and discuss formulation of media for fermentation in details. | 10 |
| C | Enumerate ideal properties of ligature and sutures. Write a brief note on catgut. | 10 |
- Q 2 Attempt any TWO of the following.**
- | | | |
|---|--|---|
| A | i) Describe the properties of an ideal plasma substitute. | 5 |
| | ii) Write a note on dextran as a plasma substitute. | 5 |
| B | i) Enumerate the risk in context to the collection, preservation and administration of the whole human blood and discuss any two in brief. | 5 |
| | ii) Write a brief note on Human Fibrinogen. | 5 |
| C | i) Discuss packaging and labeling of surgical products in general. | 5 |
| | ii) Explain the classification of bandages in brief. | 5 |

SECTION- II

- Q 3 Attempt any TWO of the following.**
- | | | |
|---|---|---|
| A | i) Explain the preparation of Polio vaccine. | 5 |
| | ii) Explain the preparation of BCG Vaccine. | 5 |
| B | i) Differentiate cellular and humoral immunity with reference to their mechanism. | 5 |
| | ii) Define the following terms: (Each carries 1 mark.) | 5 |
| | 1) Passive immunity | |
| | 2) Diagnostic aids | |
| | 3) Biosensor | |
| | 4) Opsonization | |
| | 5) Allergy | |
| C | i) Explain the preparation of Monoclonal Antibody with suitable diagram. | 5 |
| | ii) Describe physical and biochemical barriers which impart defence to human body against infection | 5 |
- Q 4 Attempt any TWO of the following.**
- | | | |
|---|--|---|
| A | i) Explain in brief about pathophysiology of allergy. | 5 |
| | ii) Explain in brief about standardization and potency of allergenic extracts. | 5 |

P.T.O.

B	i) Explain in brief about the diagnosis of allergy.	5
	ii) Answer the following questions.	
	1) Explain in brief about the use of Elisa reader.	1
	2) What is antigen- antibody reaction?	2
	3) Explain in brief about the symptoms of allergy for nasal cavity.	2
C	Write the types of biosensors and its placement in body. Write a note on any one biosensor in detail.	10
